

A Flexible Architecture for Selection and Visualization of Information in Emergency Situations

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Abstract— Emergency command and control centers (CC) are integrated facilities to assist and handle crisis situations. In these CCs, operators suffer both with information shortage and overload. This paper focuses on the information overload problem. Operators often do not have adequate access to information that may be relevant in the decision-making process. In many CC centers, information is stored without filtering or refinement for later use, and its visualization is pre-defined at design time, offering no possibility of customization. To address the problems created by this lack of support, we propose an architecture to enable fast and easy selection from multiple information sources, and definition of appropriate visualizations. Following this architecture, we developed the “Emergency Dashboard”, a system that provides selection of a desired set of data and presents it, using configurable dashboards. We also provide collaborative mechanisms, given that, at times, it is not possible for a single user to handle such large datasets alone.

Keywords— *emergency management, information visualization, dashboard, collaboration*

I. INTRODUCTION

In an emergency scenario, information comes from multiple sources and in different formats. Decision-makers can access large and diverse databases and easily obtain large amounts of information. However, part of this information may be irrelevant, which causes information overload [1]. Information overload refers to the danger of becoming lost in data, which may be irrelevant to perform a task, processed in an inappropriate way or presented in an inappropriate way [2]. In emergency scenarios, this problem remains an obstacle to effective decision making [3]. When decision-makers are overwhelmed with irrelevant information during their complex work, they can make bad decisions [4].

One of the main challenges in the development of computing environments for emergency managers is not to provide more information indiscriminately, but to provide the right information to the right person, at the right time and place, as suggested in [5]. According to [6], the difficulty of providing this information to the decision-makers lies in filtering what really matters and what should be visualized, how, for whom and when. Information overload, which slows down interpretation and decision making, must be avoided but without restricting access to information.

We have been working in collaboration with the Integrated Command and Control Center of Rio de Janeiro (CICC-RJ,

Portuguese acronym for Centro Integrado de Comando e Controle do Rio de Janeiro) since 2014. CICC-RJ is the main element in the coordination of security actions for major events in the state of Rio de Janeiro. It integrates multiple public service agencies (e.g., fire department, police, civil defense, transportation, etc.) to monitor the city and support federal, state and local agencies in establishing security, taking strategic action, and responding to emergency calls [7].

In on-site observations and interviews we perceived that emergency managers suffer with information overload: they are often overwhelmed with reports and data provided by multiple information systems that compete for their attention. They access and view the same information space, but often ignore information that may be relevant to the decision-making process. Information is stored without filtering and refinement, and visualization of information is defined at design time, both with no possibility of customization. Moreover, due to environmental dynamics, new information requirements can arise in time of use, requiring the redefinition of information of interest and the modification of the type of visualization [8].

To address this problem, we developed an architecture and the Emergency Dashboard, a system that allows a user to manipulate this influx of information. Emergency managers need to receive information in a timely manner, and it needs to be presented according to the situation's needs. This means a system to address information overload in these settings should enable emergency managers to:

- Access information from heterogeneous and distributed environments while keeping the information accessible to other emergency organizations;
- Conduct analytical operations, combining and selecting information sources according to the context;
- Visually explore the situation at hand, individually or collaboratively, by combining different types of visualizations;
- Change information and views according to the context;
- Collaborate through information and visualization exchange, share control of visualizations.

From these requirements, we have developed the Emergency Dashboard with a special focus on an intuitive

user interaction, flexibility and collaboration, as recommended by [9]. Combining these requirements, we aimed to reduce the effects of information overload and consequently improve the quality of the decisions.

This paper is structured as follows: Section 2 details the Emergency Dashboard System; Section 3 presents related works. Section 4 describes an example of scenario of study. We finalize with considerations and a discussion of future work.

II. EMERGENCY DASHBOARD

According to [9], dashboard is a visual and interactive performance management tool that displays on a single screen the most important information to achieve one or several individual and/or organizational objectives, allowing the user to identify, explore, and communicate problem areas that need corrective action.

Dashboards can support a wide range of needs for monitoring, covering all tasks that benefit from an immediate overview of what is happening, and reduce information overload [10]. A solution should simplify the full process, from the manipulation (capture and transformation) of domain data to the presentation of one or more visualizations on a

dashboard, which may involve combining multiple data streams. The proposed architecture combines existing information systems into one integrated solution at the user interface level. Figure 1 shows the Emergency Dashboard architecture, in the form of a set of interconnected components based on a client-server model. Although, this type of architecture is susceptible to network delays, since the processing is performed on the server. We believe that with the current hardware advances this kind of problem has been minimized. Components are detailed below.

A. Extractor module

The *Extractor module* is responsible for data collection and processing independent of structure. The data can come from different data sources, such as local and remote databases or social networks, among others. We have developed for each data source a content extraction process. These processes have been implemented with shell scripts or standard tools, such as Pentaho Kettle (<http://kettle.pentaho.com/>), that enables the reuse of tools developed by third parties, using software packages as needed.

This module, routes the contents extracted from data sources to the *Storage module*, following standard web formats (XML, JSON, or other).

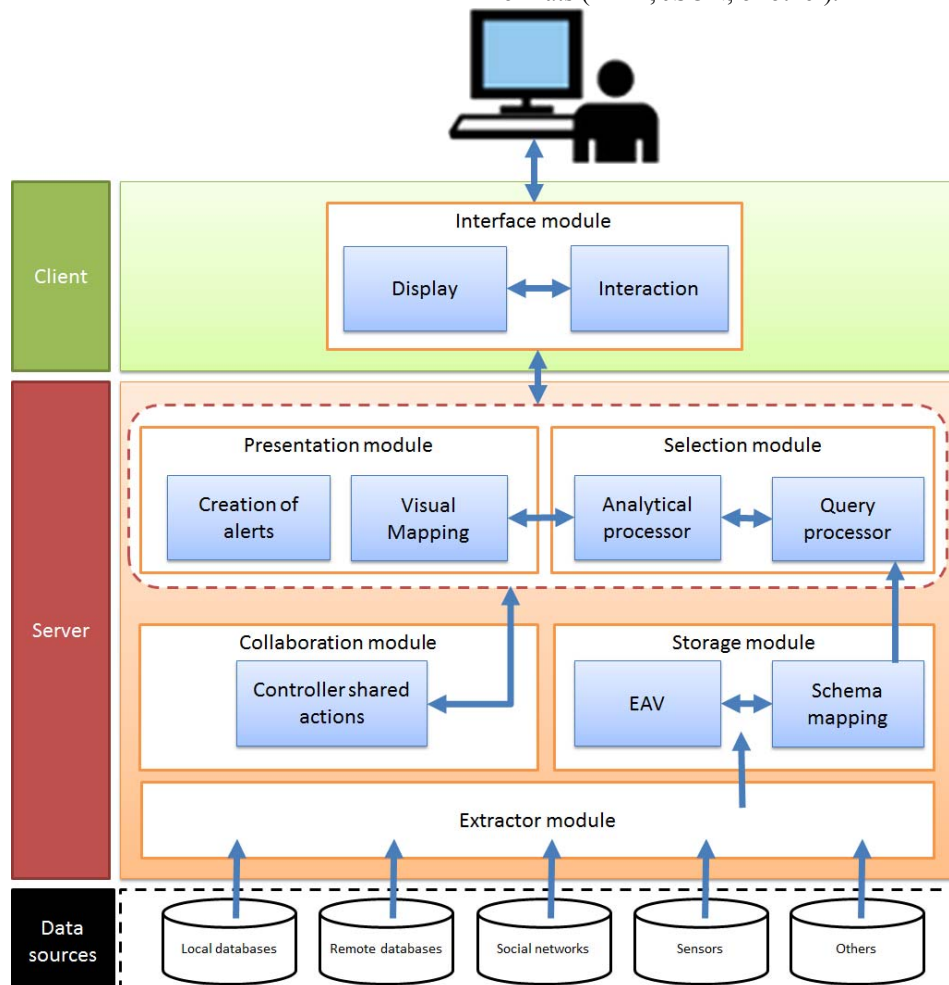


Figure 1. Emergency Dashboard Architecture

B. Storage module

The *Storage module* structures and organizes data. This module has two components: the *Entity-Attribute-Value (EAV) model* and the *Mapping Scheme*.

The EAV model, also called open schema or vertical database, is used for data modeling and can handle flexible and dynamic data structures. According to [11], in relational databases all columns (or attributes) and their types have to be specified by the application developer prior to deployment. As an alternative approach to the usual relational database, The EAV model stores data following a line-based structure. Our EAV model, as outlined in Figure 2, consist of three tables: The Entity table represents an object or event on which it wants to keep a record. The Attribute table represents the property or characteristic that describes an entity, and the Value table represents all values that are recorded. Entity and Attribute are referenced via foreign key, the value is stored as a string.

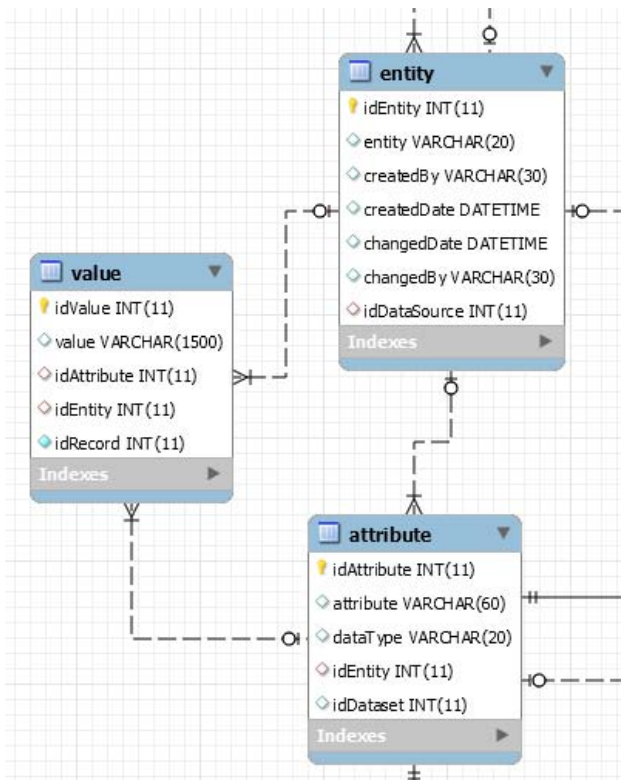


Figure 2. Part of EAV Model Design

The *Mapping Scheme* allows the user to view the original database schema. The EAV model has the *physical schema* (the way data are stored), different from the *logical schema*. It establishes the relationship between entities, their attributes and their values. Thus, this component helps the user to better interpret the logical organization of data.

C. Selection module

The *Selection module* allows the user to define what information is relevant. The first component of this module is the *Query processor*, which allows the user to view and select which information is relevant to his context. Through visual

interaction, the user can define which entities and attributes should be displayed. Since all interaction is visual, this component hides the complexity of a SQL query, which might be hard for a layperson. The result of this interaction is a table containing the selected information. The second component is the *Analytical Processor*. Given the previously generated table, this enables the user to manipulate information in a number of ways, performing summarizations, filters, removing attributes, etc. All the data manipulations are saved, making the user's dataset for later use.

D. Presentation module

The *Presentation module* performs graphics processing and generates alerts. The first component of this module is the *Visual Mapping*. This component maps information (dataset previously generated by the *Selection module*) to a visual structure. Structures may be traditional graphs (points, bar charts, pie charts or histograms), tables, more elaborate representations (using colors and geometric symbols), real images (pictures or videos), maps, diagrams (trees, graphs, networks) or visual metaphors, referring to abstract concepts and relationships. Through *Visual Mapping* the user must choose how to view his dataset. Combining elements of information (from dataset), the user might have multiple associated visual structure, and save it to a specific dashboard.

Visual structures are generated from Google's Chart API. We chose this library due to their flexibility in the use of web standards such as HTML5, CSS3 and SVG. In addition to their ability to support large datasets, a wide variety of visual structures and dynamic behavior of interaction and animation.

The second component is the *Creation of alerts*, through which managers can set alerts of events to help identify contextual changes that need attention. The alerts allow the user to define relevant events through if-then propositions. An example of alert may be defined as follows:

IF "Rainfall (mm) - Last 24 Hours" **GREATER THAN** "25" **THEN** "Flash screen".

E. Collaboration module

The *Collaboration module* is responsible for managing the collaborative aspects of the system. Through the *Controller shared actions*, it is possible to create a shared workspace with other collaborators. Results of the *Selection module*, and visual structures defined in the *Presentation module* can be manipulated by other collaborators. This mechanism collaborative can facilitate knowledge reuse, which has been found to enhance the decision making process.

F. Interface module

In the client side, we have an *Interface module* to present the result of user interactions on the dashboard. The *Display* component collects and transmits user requests to the selection and presentation modules. The *Interaction* component defines the set of actions that users can perform on the visual structures, thereby obtaining different visualizations of the same data. Using a browser, each user can setup their own dashboard.

III. RELATED WORK

Andrienko & Andrienko [12] address the intelligent visualization of situational information in Geographic Information Systems (GIS). They focus on a specific presentation for each user involved and their context. Their solution uses thematic visualization, interpretation of need and user intent and specific domain knowledge, documented in ontologies.

Wampler, Bink & Catrambone [13] presented a dynamic visualization tool called the Command Post of the Future (CPOF), which supports collaborative decision making in tactical units. The system uses a customizable workspace based on user needs, rather than a static format.

Döweling et al. [14] introduce the approach taken in the SoKNOS project to develop a new generation of emergency management systems. SoKNOS integrates multiple applications that address a large range of tasks relevant for emergency management. Those can be divided into four areas: information integration, information aggregation, information creation, and visualization and interaction.

Canós et al. [15] propose the combination of information for visualization at the command center. Previous information, contained in plans, is enriched with current contextual ones. This superimposition is called Composite Knowledge. Furthermore, the authors discuss a way to turn the display of emergency plans in navigation structures, in the form of spatial hyperlinks.

Engelbrecht et al. [6] propose a model for evaluating information regarding its usefulness for each moment. The

model was applied in a collaborative tool designed for viewing on a multi-touch table.

Ibarra et al. [16] present a strategy to select and display the "points of interest" on a digital map, according to the emergency influence area. This strategy was implemented in the Mobile-Map tool, which supports the fire service decision making in urban emergencies.

Zagorecki et al. [17] address a research on the implementation of dashboards in emergency management. The main idea is to give information about the state of the organization, through simple measures, summaries of performance and intuitive views.

A lot of research has been done on how to deal with information overload. However, we have observed, in existing work, that information and visualizations are often defined at design time. Rare are the proposals that allow customization. Our proposal differs from the aforementioned approaches, because we allow users to select information of interest and view it according their context. Each user can define their own dashboards according to their profile and context.

We propose an interactive process that enables users to redefine the information and the type of visualization on a dashboard whenever needed, regardless of the domain. In this way, through the use of dashboards, we provide relevant information in a timely manner to a decision maker as clear as possible. In addition, our approach provides for the use of collaborative mechanisms on dashboards, as shared workspace and visual structures.

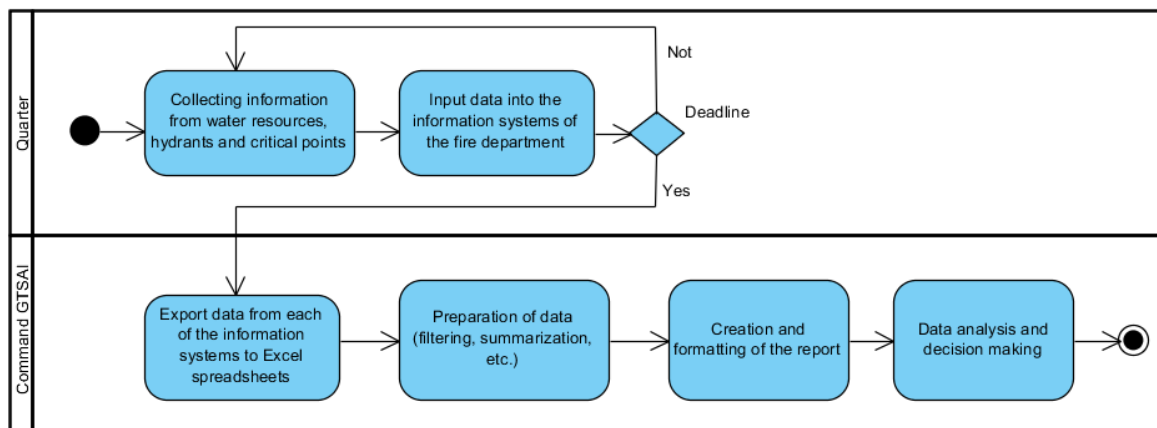


Figure 3. Area Running Scenario

IV. CASE STUDY SCENARIO

CICC-RJ is a C2 center [18], inviting key personnel who work for different government agencies, which has the capacity and autonomy to decide in a risky situation, or to resolve an incident. The main responsibility of the agents that compose the CICC-RJ is to support operational commanders by sharing information and allocating resources. In this structure, there are different agencies that compose the CICC-RJ. For this paper, we describe a real case of Fire Department. More specifically the Technical Grouping of Water Supply for

Fire (GTSAI, Portuguese acronym for Grupo Técnico de Suprimento de Água para Incêndio).

When an emergency is reported, response teams are dispatched to the location and make an assessment of the surrounding environment and resources available to deal with the situation. In many cases, the responding agency does not have information about what it will encounter at the scene of the emergency (e.g., Are there hydrants? Where are they located?) This information is crucial for the relief mission to function well.

One of GTSAI activities is to monitor the collection of water resources, hydrants, and critical points carried out by each of the quarters of Rio de Janeiro. This activity is named “Area running”. The Figure 3 shows how this activity is carried out.

During a certain period of the month the firefighter quarters are responsible for collecting information for “Area running”. Initially, all the information is collected on paper by firefighters and then are input into the information systems of the fire department. This information should be constantly updated, so that the command can extract reports (from Excel spreadsheets) to monitor the available resources and to verify whether each quarters are fulfilling their activities at the right time.

From on-site observations and interviews with two experienced managers of GTSAI. We identified several problems, such as: reports and data provided by distinct information systems that compete for their attention, loss of time in the decision-making process, lack of flexibility of reports, lack of monitoring of real-time activities, etc. Furthermore, there is a need to create different reports for different profiles or share the report with other agencies.

As a solution to this problem, we propose the use of “Emergency Dashboard”. To this end, we divide the study into two phases. First, the managers were initially trained by the authors. Through spreadsheets extracted from systems legacy, and imported into the Emergency Dashboard, managers could manipulate data sources, select relevant information and choose how the information would be presented. This phase was carried out in two meetings last approximately 80 min.

Second, in order to use real data, we developed an extraction process from legacy systems of GTSAI - VICON (Vigilance and Control – <http://www.viconsaga.com.br>) to Emergency Dashboard. From this, the managers started to redefine the information and create dashboards according to their needs.

The Figure 4 shows a dashboard created from the hydrants data. The dashboard is updated in real time and provides selective information according to the manager’s context. It is important to emphasize that the managers can modify the dashboard when necessary. They can change or add new information and visualizations. In addition, managers can set criteria-based alerts. These alerts are further contribution to help managers, as compared to the normal flow such real time monitoring were not possible.

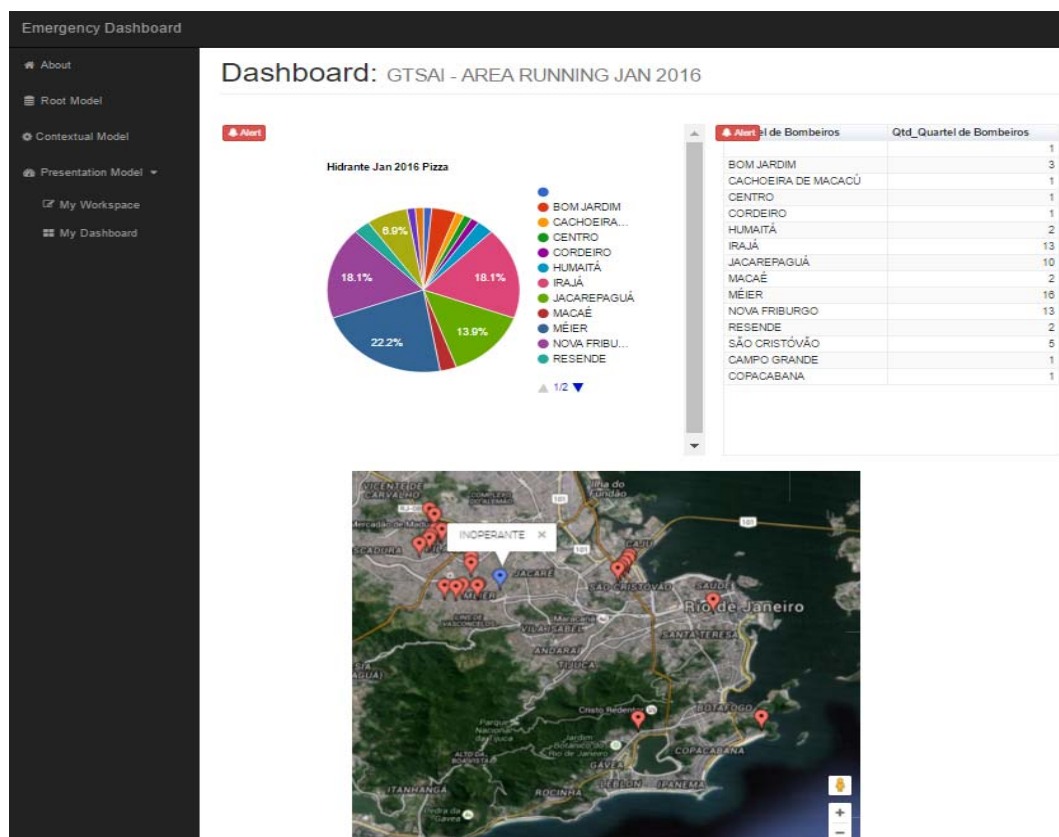


Figure 4. GTSAI Example

V. FINAL CONSIDERATIONS AND FUTURE WORK

In this paper, we propose an architecture to enable fast and easy selection from multiple information sources, and definition of appropriate visualizations interactively. From a

previous study conducted in the Integrated Command and Control Center of Rio de Janeiro (CICC-RJ), we found evidence of information overload and we verified that the success of managers depends on having the right information available, at the right time, presented in an appropriate way.

Following this architecture, we developed the “Emergency Dashboard”, a system that provides selection of a desired set of data and presents it, using configurable dashboards. We also provide collaborative mechanisms, given that, at times, it is not possible for a single user to handle such large datasets alone.

Our solution starts with the generation of domain data and ends with the presentation of one or more visualizations on a dashboard. Through a transition between models, managers can select information from raw data sources, perform some analytical operations and end with the appropriate visual representations for each type of information. The whole process is interactive, allowing the user to redefine his/her dashboard as needed.

First evaluations with users from the domain have already shown that this solution could successfully address the problem of information overload. After using the system, managers highlighted a few key elements: flexibility to redefine and add new information; focus on what is of interest; discarding information that is not relevant to their context; possibility to display maps and graphics on a single screen; control and monitoring for real-time decision making; generating alerts based on performance measurement; ability to share information with other agencies. These were considered very positive and useful by our first users.

One limitation is data integration between different information systems available. In this respect, we are considering the use of resources from the Semantic Web initiative, as suggested in [19]. In addition, we identify, in particular, collaborative dashboards as rewarding directions for future research. In order for collaborative dashboards to become used and adopted, it is important overcome several obstacles in the CCs, such as: lack of incentives at institutional, organizational and individual levels, a lack of understanding with regard to the overall operational dependencies between the various agencies, organizational norms and values, emotional reward, system usability, integration of systems in the daily routines, and information and system quality.

We are continuing this work through further evaluations with GTSAI and other agencies to identify new features that can be incorporated to solution.

Acknowledgment

Bruno S Nascimento was partially supported by grant from CAPES. Adriana S Vivacqua is partially supported by grants from 309171/2015-6 (CNPq) and E-26/202.753/2015 (FAPERJ). Marcos R.S. Borges is partially supported by grants from 484030/2013-2, and 308149/2015-7 (CNPq), and grant E-05/215.285/2015 (FAPERJ).

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